The dynamics of MM: Einstein's weak field, Friedmann's cosmology, and a boundary set by the Galaxy, with a registered test on Gaia DR4 wide binaries

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Abstract

This record gives the dynamics of Mathematical Materiality (MM). In the weak field it is linearised Einstein gravity, and the homogeneous model is Friedmann's. The junction to the $a_0 = cH_0/6$ layer is read as a boundary set by the Galaxy (the rule L*), and a premise of the author gives that boundary with two declared choices. The record also registers, before Gaia DR4 (planned for 2 December 2026), a test of this reading on wide binaries, with rules fixed before the data. It is one of MM's checks: the outcome concerns one reading of MM, not the framework. The dynamics: The prediction. In MM's reading L*, a pair of stars reads only its own field inside a boundary, where its own Newtonian field exceeds the Galactic one (7 773 AU for 1.5 solar masses near the Sun). The MOND-like boost is then the same in every direction. QUMOND with the external field predicts instead, inside that boundary, that pairs aligned with the Galactic field move more slowly. Newton predicts no difference.

The dynamics of MM: Einstein's gravity, Friedmann's cosmology and a boundary set by the Galaxy

Student version of MM record 2.70 (DOI 10.5281/zenodo.23730218), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.70 of the Mathematical Materiality (MM) series, October 2026. Mathematical Materiality, MM for short, is the author's framework of ideas. Each work of the series has a number.

How this record is made. The record has three parts.

- **Part A** is a prediction, with its rule, the computer simulations and a first look at today's data.
- **Part B** is the *dynamics* of MM, that is, its laws of motion. The prediction comes from here. It contains three things: Einstein's gravity when gravity is weak, Friedmann's equations for the *homogeneous* model, that is, the same at every point, and a boundary set by the Galaxy for the junction with the a_0 layer (section B4).
- **Part C** is a premise of the author. It gives the rule behind the prediction. And it gives a first look at what it predicts for the *amplitude*, that is, the strength of the signal, with a registered rule for the new data and an estimate of its power.

The working documents are in Italian. The archived text, in English, summarises them. The record continues MM 2.56 (DOI 10.5281/zenodo.23506401), the registered test of the amplitude of wide binaries, and it reuses its program that downloads the data.

What is put to the test. This is one of MM's checks. The outcome concerns a reading of MM, called L* (read *L-star*), not the framework.

At the end you will find how the work was done and a glossary.

Before we start: the words we need

Wide binaries. They are pairs of stars that orbit one around the other, very far from each other: thousands of times the distance between the Earth and the Sun. This distance is called the *astronomical unit* (au). The stars of a wide binary attract each other very weakly.

Gaia. It is a satellite that measures the position and the motion of a great many stars. Its data come out in blocks, called *releases*. DR3 is the release already published.

Newton's gravity and MOND. In galaxies the stars far from the centre orbit faster than expected. Visible matter alone, with Newton's law, is not enough. One answer is *dark matter*. Another is MOND, an idea of the physicist Milgrom: below a very small acceleration, a_0 (read *a-zero*), gravity is stronger than Newton's. The *acceleration* says how much the velocity changes every second.

The boost. It is the number by which Newton's gravity is multiplied. It is written ν (read *nu*). In MM the law is ν(y) = 1/(1 − e^(−√y)). Here y is Newton's acceleration divided by a_0. When y is large, ν is almost 1: Newton holds. When y is small, ν grows. In MM the scale is $a_0 = cH_0/6$: c is the speed of light and H_0 is the Hubble constant, which says how fast the universe is expanding today.

QUMOND and the external field. QUMOND is a version of MOND written with a *field equation*, which computes gravity at every point of space. In QUMOND a pair of stars also feels the gravity of the Galaxy around it: it is the *external field*. In English it is called EFE, *external field effect*.

Isotropic and anisotropic. *Isotropic* means the same in all directions. *Anisotropic* means that it changes with direction.

The words of the MM framework.

- A *premise* is a starting sentence of the author: a fixed point, like a postulate.
- A *reading* is an interpretation, written so that it can be put to the test. We also say that a body *reads* a field when its motion takes it into account.
- *Registered* means written and fixed with its date before doing the calculations. So it cannot be changed afterwards.
- The *MCD* is the greatest common divisor. An example: the MCD of 6 and 9 is 3. In the MM framework the MCD is the common unit that a relation counts.
- The *MM chain* is the row of choices of MM 2.56 for wide binaries: $a_0 = cH_0/6$, the law ν, the field of the Galaxy and QUMOND.

Two measures of how well a hypothesis works.

- The symbol σ (sigma) indicates the typical error. "2σ" means a distance of two typical errors.
- The *likelihood* says how probable a hypothesis makes the observed data. Its logarithm is used, the *log-likelihood*. The higher it is, the better.

Part A: the prediction

This part is a registration record. It was deposited before Gaia DR4, the next release of Gaia data, planned for 2 December 2026.

A1. The question

Near the Sun, a wide binary sits in the gravitational field of the Galaxy. Two readings make different predictions for the internal motion of the pair, that is, for how the two stars move with respect to each other.

reading	inside the boundary radius R_b	outside
L* (MM)	the pair reads only its own field: a MOND-like boost from its own field, **the same in every direction**	QUMOND with the external field of the Galaxy
QUMOND + external field (MOND)	the field of the Galaxy enters everywhere: the boost is **weaker along the direction of the field of the Galaxy**	the same
Newton	no boost, no dependence on direction	the same

Let us read the table.

- **L*, the reading of MM.** Inside a boundary radius R_b the pair reads only its own field. It has a MOND-like boost, the same in all directions. Outside the boundary QUMOND with the external field holds.
- **QUMOND with the external field.** The field of the Galaxy enters everywhere. The boost is weaker along the direction of the field of the Galaxy.
- **Newton.** No boost and no dependence on direction.

The boundary. R_b is the radius where the Newtonian field of the pair equals that of the Galaxy.

- For a pair of 1.5 M⊙ near the Sun, R_b = 7,773 au. M⊙ is the mass of the Sun, used as a unit.
- It is 0.861 r_M. Here r_M = √(GM/a_0) is the *MOND radius*: the distance at which the Newtonian acceleration of the pair drops to a_0. G is the gravitational constant, M the mass.
- $a_0 = cH_0/6 = 1.0914 \cdot 10^{-10}$ m s^{-2}. The unit m s^{-2} means metres per second squared. 10^{-10} means divided by ten billion.

The field. The Newtonian field of the Galaxy near the Sun is e_N = 1.3488 a_0. It comes from the speed of the Sun around the centre, V_0 = 233 km/s, and from its distance from the centre, R_0 = 8.21 kpc. A kpc (kiloparsec) is a thousand parsecs; a parsec is a little more than three light years.

What is observed. Inside the boundary two kinds of pairs are compared.

- The *aligned* pairs: the line between the two stars, seen on the sky, is close to the direction of the centre of the Galaxy (the exact rule is in section A2).
- The pairs *across*.

With L* and with Newton the aligned pairs move as fast as those across. With QUMOND they move more slowly.

What the test can separate. This test separates QUMOND from L* and Newton. The amplitude test of MM 2.56, instead, separates Newton from the other readings.

A2. What is measured and with which rule (fixed before the data)

The sample.

- They are the wide binaries of El-Badry, Rix and Heintz (2021). They had already been downloaded on 16 October 2026 for MM 2.56.
- Only the well-measured pairs are kept. The *quality cuts* were frozen on 11 October 2026: R < 0.01, RUWE < 1.4, MSMS, d < 200 pc, ϖ/σ_ϖ > 20, compatible parallaxes, 3 < M_G < 12, proper-motion error on u < 0.1, depth term < 0.1, u < 3. What the main ones mean:
- R is the probability that the two stars are close only by chance, without being a true pair;
- RUWE is an index of how well Gaia measured the star;
- MSMS means that both stars are normal stars, like the Sun;
- d is the distance from us;
- ϖ (read *varpi*) is the *parallax*, the small apparent shift that gives the distance; ϖ/σ_ϖ > 20 means that it is measured very well;
- M_G says how bright the star is.
- The masses are derived from the light of the stars (*photometric* masses).
- x_p = s/r_M. Here s is the *projected separation*: the distance between the two stars as we see it on the sky.

The velocity. u = |v_⊥|/v_N.

- v_⊥ is the part of the relative velocity on the sky perpendicular to the line between the two stars. There is no perspective correction.
- v_N = √(GM/s) is the velocity that Newton would give.
- In words: u says how fast the pair is compared with Newton's velocity.

The angle. φ (read *phi*) is the angle between the line of the pair and the direction of the centre of the Galaxy, both seen on the sky.

- The pairs are *aligned* if $|\cos\varphi| \geq 1/\sqrt{2}$, that is, if the angle between the line of the pair and the direction of the centre is at most half a right angle, in one direction or the other. Otherwise they are *across*.
- The pairs that on the sky lie within about 17° of the centre or of the *anticentre*, the direction opposite to the centre, are removed ($|\hat{e}_p| < 0.3$). Here $|\hat{e}_p|$ is the length of the direction of the centre projected on the sky: it is small when the pair lies, on the sky, almost in front of the centre or of the anticentre.

The statistic. $\Delta_\varphi = \langle u \rangle_{aligned} / \langle u \rangle_{across} - 1$. The brackets $\langle\ \rangle$ indicate the mean.

- In words: the mean u of the aligned pairs is divided by that of the pairs across, and 1 is subtracted. If the two means are equal, $\Delta_\varphi = 0$.
- It is computed in five bands of x_p. The *key band* is 0.3-0.86, inside the boundary.

The uncertainty. The *bootstrap* over the pairs is used, with 2,000 resamples: the calculation is redone many times, each time drawing at random from the pairs. No star appears in more than one pair.

The reading rule, in the key band.

- **"No QUMOND anisotropy inside the boundary"** (favours L* or Newton): if the data lie more than 2σ beyond the QUMOND interval, on the side of L*, and within 2σ of the L* interval.
- **"QUMOND-like anisotropy"** in the opposite case.
- **"Not decidable"** otherwise.

The control. In the *control band* (0.05-0.2) all readings predict from zero to 2 %. If the data there exceed 3σ, the reading carries the note "systematic in the control". A *systematic* is an error that always shifts in the same direction.

A3. The predictions (simulated before opening the data)

The models.

- Newton.
- QUMOND with the external field and the *vector force*: the part of the force along the line of the pair and the part across are computed. The calculation starts from the *phantom density*: in QUMOND the extra gravity behaves as if there were an invisible mass in empty space. A three-dimensional integral is used, for two equal masses.
- L*, with a smooth *junction* at the boundary that conserves energy: the passage from inside to outside is gradual.

The simulated population.

- 4,000 orbits, *isotropic*, that is, oriented at random.
- The *apocentre* is the point of the orbit where the two stars are farthest apart. The apocentres are distributed *log-uniformly* between 0.15 and 5 r_M: every factor of distance has the same weight.
- The *eccentricity* e says how flattened an orbit is. Its distribution is $p(e) \propto e^{1.2}$.
- The orbits are computed for 40 periods with an *adaptive leapfrog*, a calculation method with alternating steps that adapts by itself.
- Each orbit is seen from 4 lines of sight chosen at random.

The simulated measurement. It is made like that of the real data: errors on mass and distance, noise on u, fake test pairs, the same cuts.

The runs. They were computed on AWS, an online computing service, before opening the data.

band (x_p)	Newton	QUMOND+EFE	L*	expected σ with DR3
control (0.05-0.2)	[−0.006; +0.017]	[−0.006; +0.020]	[−0.005; +0.019]	0.011
inner (0.2-0.3)	[−0.007; +0.022]	[−0.014; +0.022]	[−0.007; +0.023]	0.025
key (0.3-0.86)	[−0.015; +0.007]	**[−0.031; −0.004]**	**[−0.012; +0.012]**	0.022
outer (0.86-1.5)	[−0.022; +0.006]	[+0.011; +0.035]	[+0.007; +0.030]	0.048
far (1.5-3.3)	[−0.012; +0.018]	[+0.019; +0.048]	[+0.020; +0.046]	0.082

Let us read the table. Each cell gives the predicted interval for Δ_φ.

- In the **key band** (0.3-0.86) QUMOND predicts from −0.031 to −0.004: the aligned pairs are slower. L* predicts from −0.012 to +0.012: no difference. Newton predicts from −0.015 to +0.007. The expected error with DR3 is 0.022.
- In the **control band** all three readings predict values close to zero.
- In the **outer** and **far** bands, outside the boundary, QUMOND and L* predict positive values; Newton does not.

How to read the intervals.

- They cover two time windows: periods 0-10 and 30-40. Each has its own simulation uncertainty (±).
- There is a built-in check: Newton gives zero, as it must.
- The *paired difference* L* − QUMOND in the key band is +1.1 % ± 0.4 % in periods 0-10 and +2.5 % ± 0.8 % in periods 30-40.
- The expected *power* with DR3 is 0.8σ: the difference between the readings is smaller than the error. "Not decidable" was written as the most likely outcome before the data.

A4. A first look at today's data (DR3)

Key band. There are 4,140 pairs. Δ_φ = **+0.018 ± 0.027**.

- It is 0.2σ from the L* interval, 0.4σ from Newton and 0.8σ from QUMOND.
- **Reading: not decidable with DR3.** The data do not separate the readings.

Control band. There are 17,128 pairs. The value is +0.027 ± 0.013: 2.1σ, below the 3σ threshold.

- The control band of an earlier test (F-46, August 2026, 300-2,000 au) also showed a small excess of the same sign.
- The two tests are not fully independent.
- There could be a systematic that depends on the orientation of the pairs. It has to be understood before DR4.

The spread of u. In the data u is more scattered than in the simulations. The *coefficient of variation*, that is, the spread divided by the mean, is 0.84-0.87 in the data against 0.69-0.70 in the simulations. So the real uncertainty is larger than the expected one.

A5. The rule for the new data: an exact replica

Same rule. The same pairs, re-measured in DR4, go through the same band, the same statistic, the same predictions and the same rule.

The steps. The data are downloaded; they are converted into the format of the catalogue; the pairs that in DR4 turn out to have a non-single star are removed; the sample is built; the analysis is done. The names of the programs are in the final section.

Validity. At least 3,000 pairs in the key band and 1,000 per class are needed.

The dry run on DR3. A *dry run* is a rehearsal on data already seen, to check that everything works.

- The sample builder, copied, reproduces the DR3 sample bit for bit.
- 3,000 pairs downloaded from the Gaia archive in October reproduce the same pairs, bands and classes, with u within 0.27 %.

The expected outcome. With this observable about ten times the DR3 pairs would be needed to reach 2σ. So even with DR4 the most likely outcome remains "not decidable". The record fixes the test in time.

What is not in this record. A more powerful analysis: the shape of the distribution, all the bands, the errors pair by pair. If it is done, it will be registered separately before 2 December.

A6. What the prediction rests on

The reading L*.

- A pair is read as isolated inside a sharp boundary.
- In MM this is a *rule*, not a derived law (Part B).
- Part C shows that a premise of the author, with two declared choices, gives the boundary and the isotropy inside the boundary.

The junction at the boundary.

- With a sharp switch, isotropic inside and anisotropic outside, the force does not conserve energy.
- The junction used here is smooth and conserves energy: it is a shell from 7,773 to 8,550 au. It is a choice of the author.
- How the result would change with other smooth junctions has not been measured.

An idealised population.

- In the simulations the starting distribution is known, the direction of the external field is fixed, the masses are equal in the dynamics, and there are no tides or encounters with other stars.
- In the real data the population has to be estimated, and part of the signal may be absorbed.
- The simulated numbers are *lower bounds*: the pairs needed are at least those.

Part B: the dynamics of MM behind the prediction

The starting point. MM reads gravity from counts on a *graph*, that is, a network of connected points. The closure rule is $\psi + N\psi = 2$ (ψ is read *psi*).

How it was done. Since 26 October 2026 the dynamics has been developed level by level. Each step was registered before the calculation.

B1. What holds

- **Level 0, the inventory.** The constraints of the dynamics are already in MM's *Regge-type action*. The Regge action is a way of writing Einstein's gravity on a network of flat pieces, like a ball made of many triangles.
- **Level 1.**
- The *linearised Bianchi identity* follows from the action on the lattice, with a *binding stress*. The Bianchi identity is a mathematical rule of Einstein's gravity, linked to the conservation of energy and momentum.
- If the closure is read as a law in motion, it adds a *scalar mode*, an extra kind of vibration, when the *trace* varies. The trace is a sum of components of the field.
- **The weak field.** When gravity is weak, the dynamics of MM gives *linearised* Einstein gravity: the simplified form of Einstein's equations for small deformations of space-time.
- **Level 3, cosmology.** The local *homogeneous* model, that is, the same at every point, is exactly that of Friedmann. Friedmann's equations describe how the scale of a universe that is the same at every point changes in time, whether it expands or contracts. The density $\rho(a)$ (ρ is read *rho*), that is, how the density changes with the expansion, is not fixed by the framework.

B2. Matter in motion (Level 2)

Compact stars. They are very dense stars, like neutron stars.

- A *distant reading* held fixed (D1-0) does not admit generic motions.
- A "breathing" distant reading (D1-a) gives:
- a main mode of vibration at 6.68 kHz with the matter model SLy and at 7.14 kHz with APR4. SLy and APR4 are two models of how matter is made inside neutron stars. A kHz is a thousand vibrations per second;
- a *tidal deformability* of +22 % compared with general relativity. It is how much a star is deformed under the gravity of another.
- **Open:** whether D1-a is stable is not decided.

B3. Galaxy clusters (Level 3)

A *galaxy cluster* is a very large group of galaxies bound by gravity.

- Without a_0, the dynamics of Levels 1-2 is that of Newton: a factor 6.7 of mass is missing.
- With a_0 a factor 1.5-2.2 remains, as in MOND.

B4. The junction between the closure and the a_0 layer: the open node

The problem is to connect the dynamics of MM, which gives Einstein, with the scale a_0. There were eleven attempts. First the table, then the explanation.

step	what was tried	outcome
1	algebraic hook $L = 4(\psi_0 - 1)$	not a derived junction; K1 falls on KiDS, the QUMOND realisation falls on Cassini
2	rule L*: inside R_b each system reads its own field	passes Cassini by construction; KiDS admitted; binaries with a plateau
3	local dynamics: QUMOND, a ceiling, a curvature reading	QUMOND 18.5σ and ceiling 13.7σ on Cassini (Saturn's Q_2); the curvature falls on granularity
4	AQUAL; a combination of field and curvature (D3)	AQUAL 23.5σ on Cassini; D3 passes Cassini, falls in star-poor discs
5	an extra field (AeST type), cubic Galileon	not decidable at the threshold; the tidal quadrupole grows inwards
6	L* written as a non-local law, with a rule for systems	consistent trees; two-body force validated
7	two-body QUMOND + external field in 3D	step of +6 % at the boundary relative to the isotropic mean
8	population with the mean (central) force	$N_3 = 1.3 \cdot 10^5$ pairs (3σ): "uncertain"
9	population with the vector force	$N_3 = 1\text{-}2.5 \cdot 10^4$ (idealised): the anisotropy is the main signature
10	test on DR3 (Part A)	not decidable with DR3
11, 11b, 11c	the author's premise on the observation MCD (Part C)	gives the boundary and the isotropy of L*; on DR3 its law fits best, a posteriori; DR4 is not expected to separate it from the chain

Some names, before reading the steps.

- *Cassini* is the probe that orbited Saturn. Its radio tracking limits the *quadrupole* Q_2 of the field around the Sun: a small deformation of the Sun's gravity that would change the orbit of Saturn.
- *KiDS* is a large survey of the sky that measures *weak lensing*: the gravity of galaxies bends a little the light of distant galaxies.
- AQUAL, AeST and Galileon are other theories that modify gravity.
- N_3 is the number of pairs that would be needed to see the effect at 3σ.

The steps, one by one.

1. **An algebraic hook**, $L = 4(\psi_0 - 1)$. It is not a derived junction. The reading K1 falls on KiDS. The realisation with QUMOND falls on Cassini.
2. **The rule L*:** inside R_b each system reads its own field. It passes Cassini by construction. KiDS admits it. In binaries it gives a *plateau*, a flat stretch.
3. **A local dynamics:** QUMOND, a *ceiling* on the boost, a *curvature* reading. QUMOND is at 18.5σ and the ceiling at 13.7σ on Cassini (Saturn's Q_2). The curvature falls on the test of *granularity*, another check of the series.
4. **AQUAL and a combination of field and curvature (D3).** AQUAL is at 23.5σ on Cassini. D3 passes Cassini, but falls in the discs of galaxies poor in stars.
5. **An extra field (of AeST type) and the cubic Galileon.** Not decidable at the threshold. The tidal quadrupole grows inwards.
6. **L* written as a non-local law,** with a rule for systems. *Non-local* here means that it depends on the whole system, not on the point. The *trees* of the systems (who is inside whom) are consistent, and the two-body force is validated.

7. **Two-body QUMOND with the external field, in three dimensions.** At the boundary there is a step of +6 % relative to the isotropic mean.
8. **A population of pairs with the mean, central force.** $N_3 = 1.3 \cdot 10^5$ pairs would be needed to reach 3σ: the outcome is "uncertain".
9. **A population with the vector force.** $N_3 = 1\text{-}2.5 \cdot 10^4$ pairs would be needed, in an idealised case. The main signature is the anisotropy.
10. **The test on DR3** (Part A): not decidable with DR3.
11. **The author's premise on the observation MCD** (steps 11, 11b, 11c; Part C). It gives the boundary and the isotropy of L*. On DR3 its law fits best, but a posteriori, that is, after seeing the data. DR4 is not expected to separate it from the chain.

Step 9 in more detail.

- The transverse part of the external field is not small. At the boundary $U \approx -0.16$, against ν_e $K = -0.385$ in the linear limit.
- It makes the orbits more eccentric over tens of periods. It is similar to the *Kozai cycles*, in which a distant third body gradually stretches the orbit of a pair.
- Inside the boundary L* switches it off.
- The registered corrections are six:
- a *non-commensurate* sampling, that is, without regular repetitions;
- a late time window;
- a conservative QUMOND field, derived from a *potential*. The check V8: the computed transverse part agrees within 6 % with the one derived from the potential;
- a smooth junction;
- an adaptive integrator;
- a *convergence* run, with substeps twice as fine and the energy checked over all 40 periods. It converges: $N_3 = 2.6 \cdot 10^4$ and $1.0 \cdot 10^4$, within −5 % and +1 % of the official run.

B5. In one line

- **Where the dynamics coincides with Einstein, it holds.**
- **The part proper to MM**, that is, $a_0 = cH_0/6$ and the junction with it, is still a rule, L*. It is not a derived law.
- **The local versions of the junction** tried so far fall on Cassini or on granularity.
- **L* makes a falsifiable prediction for wide binaries:** it is Part A. *Falsifiable* means that the data can disprove it.

Part C: a premise that gives L*, and what it predicts for the amplitude

C1. The premise and its translation (step 11)

The premise is the author's, of 28 October 2026. The galaxy in which a system sits acts on the *observation MCD* of the system. And this MCD, in turn, determines the system's own field.

The translation into a formula is by the AI. It has two declared choices.

- The field of the Galaxy enters as a *scale*, that is, a number, not as a vector with a direction. So the reading is isotropic.
- The boost with which a system is read cannot exceed the boost with which the Galaxy reads the point where the system is. In a formula: ν_eff = min(ν(y), ν(e_N)), that is, y_eff = max(y, e_N), with ν(y) = 1/(1 − e^(−√y)). Here *min* means the smaller of the two, *max* the larger.

This translation is called T_max.

- Two other translations, a sum and a *quadrature* (the square root of the sum of the squares), do not give a sharp boundary.
- Other choices are possible. And the reason for the minimum is not derived: "why the minimum" remains open.

What follows from it.

- **The boundary and the isotropy of L*.** The boundary falls where y = e_N: at 0.861 r_M, that is, 7,773 au for 1.5 M☉. Inside the boundary the pair reads only itself, the same in every direction. In L* boundary and isotropy were *posited*, that is, assumed; here they follow from the premise.

- **Outside the boundary T_max and L* part ways.** T_max gives an isotropic boost ν(e_N) = 1.455 and no step at the boundary. L*, which outside is QUMOND with the external field, gives 1.26-1.33, with the anisotropy.
- **The Solar System.** The anomalous acceleration is of the order of 10^{-341} m s^{-2} at 9.5 au. It is a very small number, far below any measurement. T_max passes.
- **The globular cluster NGC 2419.** A *globular cluster* is a ball of a great many stars. Here T_max gives a boost ν ≈ 3.3-3.4 at 86 pc, while the data give at most about 1.1 ($\lesssim$ 1.1). **T_max falls**, as the book finds for every law made of the internal and external accelerations alone. This outcome hits the translation T_max. The book's way out is the history of the cluster, which the premise does not replace.

C2. T_max against today's data, with the machinery of MM 2.56 (step 11b)

Registered before the calculation.

- The *orbit libraries* for T_max: 200,000 orbits each, computed with the integrator of MM 2.56. Only the force changed.
- The *forward simulations*, with the *hidden companions* (third stars that cannot be seen) and the Poisson likelihood on 26 classes per band, as in MM 2.56.
- The same thresholds as MM 2.56: 5, and 2 for each *subset* with one replica removed.
- A sensitivity variant (2c), with the two-body reading of step 6 inside the boundary.

The checks.

- The libraries pass the criteria of MM 2.56: the energy error at the 99th percentile is $9\cdot10^{-4}$, and no pair starts unbound. The 99th percentile means that 99 per cent of the orbits have a smaller error.
- With its six hypotheses, the evaluation reproduces MM 2.56 without changes (D = +7.64).

The DR3 development data. They are the El-Badry pairs: 913 in the low-acceleration band, and 4,000 simulated pairs in the control band. Here is the log-likelihood of each hypothesis, counted from the best one.

hypothesis	ΔlogL
T_max	**0**
T_max, two-body variant	−0.6
MOND, simple	−1.1
MOND, standard	−1.9
MM chain 674 (QUMOND, $a_0 = cH_0/6$; L* outside the boundary)	−3.4
MM chain 730	−5.3
MLS, $a_0 = 1.2\cdot10^{-10}$	−9.8
Newton	−11.1

Let us read the table. ΔlogL is the difference from the best hypothesis: 0 for the best, negative numbers for the others.

- T_max is the best (0). Its two-body variant gives −0.6.
- MOND in the *simple* form gives −1.1, in the *standard* form −1.9. They are two ways of writing the passage between Newton and MOND.
- The MM chain gives −3.4 in the version called 674 and −5.3 in the one called 730. Chain 674 uses QUMOND with $a_0 = cH_0/6$, and L* outside the boundary.
- MLS, that is, the law ν with $a_0 = 1.2\cdot10^{-10}$, gives −9.8.
- Newton gives −11.1.

The outcome with the registered criterion.

- T_max against the MM chain: +3.4, with subsets 0.2, 9.1 and 5.3. **Not decidable.**

- T_max against Newton: +11.1, with subsets 7.8, 13.6 and 7.4. **T_max favoured.**
- The registered two-body variant, with equal masses, gives the same classes. This does not show that the result does not depend on every other way of treating the two bodies.

Where the difference comes from. Almost all of it from the low-acceleration band. There T_max is an isotropic boost of 1.455.

A posteriori. The idea of T_max outside the boundary came partly from these same data. The best MOND hypothesis of MM 2.56 has a boost of about 1.45 there. On DR3 this is *coherence*, not a test. And the differences of 1-2 between the MOND-like hypotheses lie within the *noise* of about ±3 seen in MM 2.56: the noise is how much the result changes by pure chance in the simulations.

C3. The rule for the new data

The chain of steps. It is the same as for MM 2.56: the data are downloaded, they are converted, and the forward simulations of its six hypotheses are made. Then the simulations with the two registered libraries of T_max and their evaluation are added. The new rule is added to that of MM 2.56 without changing it. The names of the programs are in the final section.

Frozen with the rule.

- 4,000 simulated pairs in the control band, as in MM 2.56.
- The thresholds 5 and 2. They are a practical criterion registered beforehand. They are not calibrated with simulations of *false positives*, that is, of cases in which the test would favour the wrong hypothesis.

DR4 is not an independent replica of DR3. The pairs are largely the same, re-measured with more data. The test is *prospective*, that is, it looks ahead, because the rule is fixed before the data.

The expected outcome: not decidable (step 11c, registered before the calculation). Mock data were made with the same machinery: 40 mock experiments for each true hypothesis, with from 1 to 8 times the DR3 pairs of the low-acceleration band.

- With the DR3 pairs, as in the DR4 replica, the rule picks the right reading with probability 0.12 (5/40) if T_max is true, and 0.03 (1/40) if the chain is true. In one sentence: "with the same pairs DR4 probably does not separate them".
- With 8 times the pairs the probability is still only 0.38 and 0.05: the advantage does not grow. The targets of 50 % and 80 % are not reached within 7,304 simulated pairs. Beyond that number the threshold is not estimated.
- When the chain is true, the criterion favours T_max by mistake in 3 experiments out of 40 at the size of DR3 (0.07-0.12 at the various sizes). It is a *Monte Carlo* estimate, that is, made with random simulations, very uncertain.

A compatible explanation. It is an a posteriori *diagnostic*, that is, a check to understand why, registered before running it.

- T_max and the chain differ mainly in the amplitude of the boost.
- A change of amplitude can be absorbed by the *mass scale*, which the rule leaves free and common to the two bands.
- With 8 times the pairs each reading moves towards the other by changing the masses by about 7 %.
- With the mass scale fixed exactly at the true value, 8 times the pairs give 0.82 and 0.65. It is a favourable case, not a requirement. Which precision on the masses would be enough is not simulated.
- The diagnostic does not isolate the cause. Two other causes are not excluded: the Monte Carlo noise of the models of the control band, fixed at the three registered replicas, and the coarse calculation grids.

The +3.4 seen on DR3 does not choose between the two readings. In a *Gaussian* approximation of the simulated distributions (bell-shaped), the ratio of the densities is about 1.1. It is a description made a posteriori.

Two checks of T_max with DR4.

- The amplitude, with this rule.
- The direction, with Part A. Inside the boundary: no anisotropy, as for L*. Outside the boundary T_max predicts none, while L* and QUMOND predict from +0.7 % to +4.8 %. With this observable those bands are not decidable.

C4. What it means

- **The premise, with two declared choices, turns the rule L* into a consequence** inside the boundary. And it adds a prediction outside the boundary.
- **On DR3, a posteriori,** its law fits best of the eight hypotheses. But it is not distinguishable from the MM chain.
- **With DR4 and the registered rule** the expected outcome is again "not decidable" (step 11c). The amplitude test separates the two MOND-like readings from Newton. It hardly separates them from each other.
- **The outcomes concern the translation T_max,** not the premise nor MM.

How the work was done

This section collects the procedure. In the archived text it corresponds to the sections Verification and Contents.

Who checked. Each step was checked by two AI verifiers, "Astra" (Codex, GPT-6) and "Argon" (Gemini), in one or two rounds. Their reports and the author's answers are in the `verification/` folders. For step 9 the last round left one point open: the convergence of the official run. Correction 6, which converges, answered it.

Who did the calculations. The author is not a professional physicist. The calculations were done with AI assistants (Claude, checked by Codex and Gemini). No human *referee* has reviewed them. A referee is an expert who checks a work before publication.

The registrations. For Part B, the fingerprints and the UTC times are in `registers/`; the notes are in `B_dynamics/notes/`, in Italian. A SHA-256 *fingerprint* is a code computed from the content of a file: if the file changes even slightly, the fingerprint changes. The premise of Part C, in Italian, is in `C_premise/notes/PREMESSA_UGO_20261028.md`. The translation T_max is by the AI that wrote the archived text.

The programs of Part A, rule for the new data.

- download: `scarica_dr4.py` of MM 2.56, mode `dr4`, fallback `dr4_posizione`;
- `code/dr4/converti_p10.py`: from the format of the Gaia archive to the format of the catalogue; it removes the pairs with a DR4 solution that is not a single star;
- `costruisci_p10_dr4.py`;
- `analisi_p10_dr4.py`.

The programs of Part C, rule for the new data. After the chain of MM 2.56, `esegui_p11b_dr4.sh`: `forward_tmax.py` for the two registered libraries and `valuta_tmax_c1.py` in mode `dr4`. The convergence correction of step 9 is described in NOTA_ESITO_PASSO9_v3.md.

What is in the package.

- **A_prediction/documents:** protocol, predictions, outcome note and rule for DR4, in Italian.
- **A_prediction/code:** sample builder, simulation and analysis; dr4/ contains the programs of the replica.
- **A_prediction/data:** the sample with the Gaia identifiers; the query and the *checksums* of the El-Badry download.
- **A_prediction/results:** simulations (npz, json, log), analysis of DR3, dry run for DR4.
- **B_dynamics/notes:** final notes of Levels 0-3 and of steps 1-9 of the junction, including the protocol and the corrections of step 9.
- **B_dynamics/code** and **B_dynamics/results:** the codes and the outputs of the binary chain (steps 6-9).
- **C_premise:** the premise, the protocols, the notes, the code, the outputs and the verification of steps 11, 11b and 11c. The mock models and the experiments of step 11c are in `results/p11c_modelli` and `results/p11c_pseudo`.
- The two orbit libraries (217 MB each) are not included. They are rebuilt with `esegui_p11b.sh`, from the *seeds* 17 and 18, in about 10 minutes each. The seed is the starting number of the random numbers. Their SHA-256 are in `registers/`.
- **registers:** the REGISTRO.txt of each working folder, with the SHA-256 and the UTC registration time of each file.
- **IMPRONTE.sha256:** the checksums of all the files of the record.

The paths. The programs use the paths of the author's computer. The El-Badry download (167,301 pairs, 138 MB) is not included; the query and the checksums are.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a

full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

The words of the framework (MM)

- **L* (L-star).** The reading of MM in which a pair, inside a boundary, reads only its own field, the same in every direction. In MM it is a rule, not a derived law.
- **MCD.** The greatest common divisor. In the MM framework it is the common unit that a relation counts. The *observation MCD* is the one with which a system is observed.
- **MM chain.** The row of choices of MM 2.56 for wide binaries: $a_0 = cH_0/6$, the law ν, the field of the Galaxy and QUMOND.
- **Premise.** A starting sentence of the author, a fixed point like a postulate.
- **Reading.** An interpretation, written so that it can be put to the test. A body *reads* a field when its motion takes it into account.
- **Registered.** Written and fixed with its date before the calculations, so that it cannot be changed afterwards.
- **T_max.** The translation into a formula of the premise of 28 October 2026: the boost of a system does not exceed the one with which the Galaxy reads the point where it is.

The words of physics

- **a_0.** A very small acceleration: below a_0, in MOND, gravity is stronger than Newton's. In MM $a_0 = cH_0/6$.
- **A posteriori.** Said of a check made after seeing the data.
- **Acceleration.** How much the velocity changes every second.
- **Anisotropic, isotropic.** That changes with direction; the same in all directions.
- **au (astronomical unit).** The mean distance between the Earth and the Sun.
- **Boost (ν).** The number by which Newton's gravity is multiplied.
- **Bootstrap.** A way of estimating the uncertainty by redoing the calculation many times on samples drawn at random from the data.
- **Boundary (R_b).** The radius where the Newtonian field of a pair equals that of the Galaxy.
- **Dry run.** A rehearsal on data already seen, to check that everything works.
- **Eccentricity.** How flattened an orbit is.
- **External field.** The gravity that comes from outside a system; for binaries near the Sun it is that of the Galaxy.
- **Gaia, DR3, DR4.** Gaia is a satellite that measures positions and motions of stars; DR3 is the data release already published, DR4 the next one.
- **Likelihood.** How probable a hypothesis makes the observed data.
- **Log-likelihood.** The logarithm of the likelihood: the higher it is, the better a hypothesis accounts for the data.
- **MOND, QUMOND, AQUAL.** MOND is Milgrom's idea: below a_0 gravity is stronger. QUMOND and AQUAL are two versions written with a field equation.
- **MOND radius (r_M).** The distance at which the Newtonian acceleration of a body drops to a_0.
- **Monte Carlo.** A calculation made with many random simulations.
- **Phantom density.** In QUMOND, the invisible mass that seems to sit in empty space.
- **Projected separation (s).** The distance between the two stars of a pair as we see it on the sky.
- **Sigma (σ).** The typical error of a measurement.
- **Systematic.** An error that always shifts in the same direction.
- **Wide binaries.** Pairs of stars very far from each other that orbit one around the other.

References

Works cited in the record, with their DOI.

- A (1995). Thermodynamics of Spacetime: The Einstein Equation of State. https://doi.org/10.1103/physrevlett.75.1260
- A (2016). Entanglement Equilibrium and the Einstein Equation. https://doi.org/10.1103/physrevlett.116.201101
- Abbott et al. (2017). Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A. https://doi.org/10.3847/2041-8213/aa920c
- Abbott et al. (2017). GW170814: A Three-Detector Observation of Gravitational Waves from a Binary Black Hole Coalescence. https://doi.org/10.1103/physrevlett.119.141101
- Abbott et al. (2017). GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral. https://doi.org/10.1103/physrevlett.119.161101

- Abbott et al. (2017). Search for Post-merger Gravitational Waves from the Remnant of the Binary Neutron Star Merger GW170817. https://doi.org/10.3847/2041-8213/aa9a35
- Abbott et al. (2018). GW170817: Measurements of Neutron Star Radii and Equation of State. https://doi.org/10.1103/physrevlett.121.161101
- Abbott et al. (2019). Tests of General Relativity with GW170817. https://doi.org/10.1103/physrevlett.123.011102
- Abbott et al. (2019). Properties of the Binary Neutron Star Merger GW170817. https://doi.org/10.1103/physrevx.9.011001
- Abbott et al. (2020). GW190425: Observation of a Compact Binary Coalescence with Total Mass $\sim 3.4\ M_{\odot}$. https://doi.org/10.3847/2041-8213/ab75f5
- Abbott et al. (2021). Tests of general relativity with binary black holes from the second LIGO-Virgo gravitational-wave transient catalog. https://doi.org/10.1103/physrevd.103.122002
- Abbott et al. (2025). Tests of general relativity with GWTC-3. https://doi.org/10.1103/physrevd.112.084080
- Adhikari et al. (2021). Probing Galaxy Evolution in Massive Clusters Using ACT and DES: Splashback as a Cosmic Clock. https://doi.org/10.3847/1538-4357/ac0bbc
- Aker et al. (2022). Direct neutrino-mass measurement with sub-electronvolt sensitivity. https://doi.org/10.1038/s41567-021-01463-1
- Aker et al. (2025). Direct neutrino-mass measurement based on 259 days of KATRIN data. https://doi.org/10.1126/science.adq9592
- Andersson et al. (1998). Towards gravitational wave asteroseismology. https://doi.org/10.1046/j.1365-8711.1998.01840.x
- Angus et al. (2008). X-ray group and cluster mass profiles in MOND: unexplained mass on the group scale. https://doi.org/10.1111/j.1365-2966.2008.13353.x
- Arnowitt et al. (2008). Republication of: The dynamics of general relativity. https://doi.org/10.1007/s10714-008-0661-1
- Bertotti et al. (2003). A test of general relativity using radio links with the Cassini spacecraft. https://doi.org/10.1038/nature01997
- Biviano et al. (2013). CLASH-VLT: The mass, velocity-anisotropy, and pseudo-phase-space density profiles of thez= 0.44 galaxy cluster MACS J1206.2-0847. https://doi.org/10.1051/0004-6361/201321955
- Biviano et al. (2023). CLASH-VLT: The Inner Slope of the MACS J1206.2-0847 Dark Matter Density Profile. https://doi.org/10.3847/1538-4357/acf832
- Cao et al. (2017). Space from Hilbert space: Recovering geometry from bulk entanglement. https://doi.org/10.1103/physrevd.95.024031
- Cao et al. (2018). Bulk entanglement gravity without a boundary: Towards finding Einstein’s equation in Hilbert space. https://doi.org/10.1103/physrevd.97.086003
- Chang et al. (2018). The Splashback Feature around DES Galaxy Clusters: Galaxy Density and Weak Lensing Profiles. https://doi.org/10.3847/1538-4357/aad5e7
- Diaferio (1999). Mass estimation in the outer regions of galaxy clusters. https://doi.org/10.1046/j.1365-8711.1999.02864.x
- Eckert et al. (2018). Non-thermal pressure support in X-COP galaxy clusters. https://doi.org/10.1051/0004-6361/201833324
- Ettori et al. (2018). Hydrostatic mass profiles in X-COP galaxy clusters. https://doi.org/10.1051/0004-6361/201833323
- Everitt et al. (2011). Gravity Probe B: Final Results of a Space Experiment to Test General Relativity. https://doi.org/10.1103/physrevlett.106.221101
- Famaey et al. (2012). Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions. https://doi.org/10.12942/lrr-2012-10
- Flanagan et al. (2008). Constraining neutron-star tidal Love numbers with gravitational-wave detectors. https://doi.org/10.1103/physrevd.77.021502
- Gourgoulhon (2007). 3+1 Formalism and Bases of Numerical Relativity. https://doi.org/10.48550/arxiv.gr-qc/0703035
- Gunn et al. (1972). On the Infall of Matter Into Clusters of Galaxies and Some Effects on Their Evolution. https://doi.org/10.1086/151605
- Hinderer (2008). Tidal Love Numbers of Neutron Stars. https://doi.org/10.1086/533487
- Hinderer (2009). ERRATUM: “TIDAL LOVE NUMBERS OF NEUTRON STARS” (2008, ApJ, 677, 1216). https://doi.org/10.1088/0004-637x/697/1/964
- Ibata et al. (2011). THE GLOBULAR CLUSTER NGC 2419: A CRUCIBLE FOR THEORIES OF GRAVITY. https://doi.org/10.1088/0004-637x/738/2/186
- Israel et al. (2005). The Discovery of Rapid X-Ray Oscillations in the Tail of the SGR 1806-20 Hyperflare. https://doi.org/10.1086/432615
- Kramer et al. (2021). Strong-Field Gravity Tests with the Double Pulsar. https://doi.org/10.1103/physrevx.11.041050
- Merten et al. (2015). CLASH: THE CONCENTRATION-MASS RELATION OF GALAXY CLUSTERS. https://doi.org/10.1088/0004-637x/806/1/4

- More et al. (2015). THE SPLASHBACK RADIUS AS A PHYSICAL HALO BOUNDARY AND THE GROWTH OF HALO MASS. https://doi.org/10.1088/0004-637x/810/1/36
- More et al. (2016). DETECTION OF THE SPLASHBACK RADIUS AND HALO ASSEMBLY BIAS OF MASSIVE GALAXY CLUSTERS. https://doi.org/10.3847/0004-637x/825/1/39
- Oppenheimer et al. (1939). On Continued Gravitational Contraction. https://doi.org/10.1103/physrev.56.455
- R. et al. (2004). The Dynamics of General Relativity. https://doi.org/10.48550/arxiv.gr-qc/0405109
- Rines et al. (2013). MEASURING THE ULTIMATE HALO MASS OF GALAXY CLUSTERS: REDSHIFTS AND MASS PROFILES FROM THE HECTOSPEC CLUSTER SURVEY (HeCS). https://doi.org/10.1088/0004-637x/767/1/15
- Sanders (1999). The Virial Discrepancy in Clusters of Galaxies in the Context of Modified Newtonian Dynamics. https://doi.org/10.1086/311865
- Sanders (2003). Clusters of galaxies with modified Newtonian dynamics. https://doi.org/10.1046/j.1365-8711.2003.06596.x
- Sanders (2011). NGC 2419 does not challenge modified Newtonian dynamics. https://doi.org/10.1111/j.1745-3933.2011.01160.x
- Sartoris et al. (2020). CLASH-VLT: a full dynamical reconstruction of the mass profile of Abell S1063 from 1 kpc out to the virial radius. https://doi.org/10.1051/0004-6361/202037521
- Shao et al. (2012). New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars. https://doi.org/10.1088/0264-9381/29/21/215018
- Shin et al. (2019). Measurement of the splashback feature around SZ-selected Galaxy clusters with DES, SPT, and ACT. https://doi.org/10.1093/mnras/stz1434
- Shin et al. (2021). The mass and galaxy distribution around SZ-selected clusters. https://doi.org/10.1093/mnras/stab2505
- Skordis et al. (2021). New Relativistic Theory for Modified Newtonian Dynamics. https://doi.org/10.1103/physrevlett.127.161302
- Strohmayer et al. (2006). The 2004 Hyperflare from SGR 1806−20: Further Evidence for Global Torsional Vibrations. https://doi.org/10.1086/508703
- Tian et al. (2020). The Radial Acceleration Relation in CLASH Galaxy Clusters. https://doi.org/10.3847/1538-4357/ab8e3d
- Tinto et al. (2014). Time-Delay Interferometry. https://doi.org/10.12942/lrr-2014-6
- Umetsu et al. (2016). CLASH: JOINT ANALYSIS OF STRONG-LENSING, WEAK-LENSING SHEAR, AND MAGNIFICATION DATA FOR 20 GALAXY CLUSTERS*. https://doi.org/10.3847/0004-637x/821/2/116
- Weisberg et al. (2016). RELATIVISTIC MEASUREMENTS FROM TIMING THE BINARY PULSAR PSR B1913+16. https://doi.org/10.3847/0004-637x/829/1/55